# SPECIFICATIONS
# FOR
# MAIN DIESEL GENERATOR ENGINE

(550 kW ACG x 900 $min^{-1}$ 3 sets/ship)

DAIHATSU DIESEL MFG.CO. LTD.
TECHNICAL DEPARTMENT
OSAKA JAPAN

| | | | |
|---|---|---|---|
| HULL NO. | DB66-01/02 C | DRAWN BY | |
| ENG. MODEL | 6DE-18 | CHECKED BY | Y. Shimomura |
| LIST NO. | AQA10036411E | | |
| DATE | March 29, 2022 | APPROVED BY | |
| REVISION | B:P11/13 2022/06/27 S.I<br>C:P1/13 2022/11/02 S.I<br>D:P2/13 2023/01/10 S.I<br>E:P4/13 2023/09/11 S.I | | |

## 1. GENERAL

1.1 Rules and Regulations

The diesel generator engine will be complied with the requirements of the following rules and regulations of the latest issue.

1) Class..... NK, M0
2) Compliance for Engine Air Pollution Prevention with Marpol, AnnexVI, Regulation 13
Statement of Compliance issued by NK M0 should be submitted. IMO Tier III Imitation of MARPOL 73/78 ANNEX VI
3) Japanese industrial standard .....JIS
4) Miscellaneous
The other not specifically mentioned in this specifications will be manufactured on the basis of manufacturer's quality control standard.
5) Flag..... PANAMA D

1.2 Conditions

| | |
|---|---|
| Machinery ..... | ambient temp. +5 to 45°C<br>cooling fresh water temp. 36°C<br>relative humidity 85 %<br>barometric press. 1013 hPa (760mmHg) |
| General power source ..... | AC 440 V 60Hz 3-phase |
| Control power source ..... | AC 220 V 60Hz 1-phase and DC 24 V |
| Starting air source ..... | 3.0 MPa |
| Control air source ..... | 0.7 - 0.9 MPa |
| Heating steam source..... | 0.8 MPa saturated, working |

1.3 Installations

Diesel engine will be directly coupled to the generator as maker's standard and placed together on the common bed.
Diesel generator set should be rigidly installed onto the hull structure as maker's standard.

1.4 Pipe connection flanges

Pipe connection flanges which are connected to the shipyard's piping will be provided in accordance with JIS.
JIS counter-flanges will be supplied by Daihatsu.

1.5 Name and caution plates

The plates will be written in English.
The caution plates will be written in English.
Unit is the SI system. (i.e. pressure unit is MPa and temperature unit is °C )

1.6 Painting

After coating with anti-corrosive paint in accordance with manufacturer's standard, the finished color will be painted as follows.

| | |
|---|---|
| Generator engine and accessories..... | Munsell 7.5BG7/2 |
| Exhaust manifold cover ..... | Aluminum heat resisting paint |
| Control panel ..... | Munsell 7.5BG7/2 |

1.7 Plans

Approval plan.....5 copies + PDF
Working plan.....5 copies + PDF
Final plan.....5 copies+ soft copy of final drawings, instruction manuals and test report for each ship
1 Original of certificate for each ship

1.8 Fuel oil

Heavy Fuel Oil (hereinafter referred to as HFO) ..... ISO 8217-F-RMG700
of viscosity up to 700mm2/s(cSt)
at 50°C

Marine Diesel Oil (hereinafter referred to as MDO).. ISO 8217-F-DMB or better grade,MGO

Low Sulphur oil(0.1%m/m)(hereinafter referred to as LSFO)...only use at in ECA.

1.9 Notes for Fuel oil pre-treatment

1) Upon deep deliberation with separator maker, the best centrifugal separation system should be applied.
2) Recommended fuel properties before engine

| | |
|---|---|
| Aluminum content | : < 10 mg/kg |
| Water content | : < 0.2 % by volume |
| Sodium content | : < 50 mg/kg |

1.10 Fuel oil when engine starting, stopping and normal condition

Engine starting and stopping .....MDO/HFO(initial start and cold condition: MDO)

Normal operation .....HFO

Note: In case that the engine is started on HFO, the standby engine needs to be pre-heated up with the circulating jacket water of 70°C.

1.11 Notices for operation

The engine, together with proper external system arrangements, can be operated on the above-mentioned fuel, still:

- viscosity of HFO at the engine inlet port should be kept to 11~14mm2/s(cSt) in any case.
- viscosity of MDO at the engine inlet port should be kept to≥2.0mm2/s(cSt)
  (With any MDO viscosity, please keep 2.0mm2/s at engine inlet port.)
- recommendable continuous low load ≥ 20% (110kW Gen.output) in case that CCAI(Calculated of Carbon and Aromatics Index) should be no more than 850.
- no load operation on HFO before stopping is proper to as short as possible, less than 5 min.

1.12 Recommended lubricating oil

| | | | |
|---|---|---|---|
| System oil..... | API service grade CD | SAE30or40 | TBN30 to 40 |
| Governor oil..... | same as system oil | SAE40or40 | |

1.13 Fluid holdings in engine

Lubricating oil.....LO tank abt.1,100lit = initial filling quantity about 1,160lit
(engine in running condition 65lit)

Governor oil..... 1.3lit

Engine CW..... 160lit

## 2. PRINCIPAL PARTICULARS

2.1 Type of Engine ..... Vertical in-line 4-stroke single-acting non-reversible direct injection trunk piston type with exhaust turbocharger and air cooler

2.2 Type of Generator ..... synchronous self-exciting brushless self-ventilated drip-proof bracket type self-lubricating one(1) sleeve bearing
Supplier: DAIHATSU
Maker:NISHISHIBA

2.3 Direction of rotation ..... clockwise as viewed from generator end

2.4 Control ..... manual start/stop : remote and engine-side
automatic start : managed by MSB
automatic stop : caused by engine emergency shutdown
synchronization : manual/auto on MSB

2.5 Engine starting method ..... Compressed air start

2.6 Engine cooling system ..... central cooling system with max 38°C LTFW

2.7 Location of Turbocharger ..... opposite side to generator

2.8 Camshaft ..... built up type

2.9 Diesel generator data

| | |
|---|---|
| Engine model | 6DE-18 |
| Quantity of cylinders | 6 |
| Cylinder bore x Piston stroke | 185 mm x 280 mm |
| Engine rated output | 600 kW |
| Engine speed | 900 min-1 |
| Mean effective pressure | 1.87 MPa |
| Mean piston speed | 8.40 m/s |
| Maximum pressure | ≦20.0 MPa |
| Overload capacity | 110% for an hour every 12 hours |
| Lube oil consumption | 0.8g/kWh (estimate at full load) |
| Fuel oil consumption *1)*2) | abt.194 g/kWh +5% with pumps(Tier II) |

| | |
|---|---|
| Generator capacity | 687.5 kVA |
| Generator rated output | 550 kW |
| Power factor | 0.8 lagging |
| No. of poles | 8 |
| Voltage Phase Frequency | AC 450 V 3-phase 60 Hz |
| Insulation | Class F |

| | |
|---|---|
| Quantity of DG set | Three(3) sets/ship |

*1)Fuel oil consumption is based on the engine output (kW) at crankshaft-end using fuel lower calorie of 42700kJ/kg, under 100% generator load, on condition that the with engine drives each one (1) of LO pump and two (2) CW pumps.
The shop trial will be carried out test on this figure. E

*2)Fuel oil consumption is based on **IMO Tier 2 limitation** of MARPOL 73/78 ANNEX VI

## 3. SPEED GOVERNING SYSTEM

3.1 Engine fitting (each 1 set/engine)

| | |
|---|---|
| Governor | mechanical-hydraulic type<br>Bosch Automatic systems corporation made<br>model RHD6-MC |
| Governor motor | permanent magnetic field type DC24V motor<br>speed changing time: 7-10 s/Hz |

3.2 Separate accessory (installed on hull part by shipyard):

| | |
|---|---|
| Governor speed control unit | potentiometer and rectifier<br>input from MSB: AC200-220V 60Hz 1-phase<br>output to governor motor: DC16-24V |

3.3 Reference data for External system

| | |
|---|---|
| Sudden load steps | 0→50→100%: 2 steps |
| Sudden off-load step | 100→0% |
| Speed variations | within 10% momentary and within 5% permanently |
| Speed stabilized time | within 5 seconds |

## 4. CHARGE AIR / EXHAUST SYSTEM

4.1 Engine fitting (each 1 set/engine)

| | |
|---|---|
| Exhaust turbo-charger | MHI made model MET18SRC<br>without cleaning device of compressor and turbine side |
| Air cooler | tube with fin type fresh water cooled |

4.2 Separate accessory (installed on hull part by shipyard)

| | |
|---|---|
| Exhaust expansion joint | for exhaust gas turbocharger outlet<br>stainless steel bellows with both end flanges<br>size 250A x 350A (1 pc./engine) |
| Exhaust gas silencer | vertical type without spark arrester 15dB(A)<br>size 350A (1pc./engine) |

4.3 Reference data for External system

| | |
|---|---|
| Fresh air outlet from ventilator | to be located near the turbocharger air inlet |
| Required combustion air | 4,300 $m^3$/h at 25°C per one(1) engine |
| Charge air temp. before cylinders | operation 45 - 55°C |
| Exhaust gas volume | 4,050 $m^3$ normal/h per one(1) engine |
| Exhaust gas temp. at T/C outlet | approx.300°C |
| Permissible exhaust gas back press. | max.550 mmAq (with SCR) |

Remark: In case that exhaust back press. over the permissible limit, such as apply the exhaust gas treatment system, 550mmAq is permissible with FOC increase.

## 5. FUEL OIL SYSTEM

5.1 Engine fitting (each 1 set/engine)

| | |
|---|---|
| F.O. final filter | notch wire duplex type with change-over handle |
| (at engine inlet) | 200 mesh (absolute 75 micron) |
| F.O. high-pressure pipe | block type |
| F.O. level switch box | leaked oil detect |
| F.O. Press.relief valve | at oil return pipe end Δp=0.1 - 0.2MPa |
| F.O. piping heat treatment | Laging & steam trace |
| Pressure damper | on oil inlet and outlet pipe |

5.2 Separate accessories (installed on hull part by shipyard): None

5.3 Reference data for External system

| | | |
|---|---|---|
| MDO service tank height | low alarm level is to be higher than the cylinder cover top of engine | |
| HFO pre-treatment with centrifugal separator | Aluminum content | : <10mg/kg |
| | Water content | : <0.2% by volume |
| | Sodium content | : <50mg/kg |
| MDO feed pump | ≥0.50m³/h per 1 engine discharge: 0.8MPa, suction: static | |
| HFO pre-boost (supply) pump | ≥0.20m³/h pre 1 engine discharge: 0.4MPa, suction : static | |
| HFO booster (circulation) pump | ≥0.55m³/h pre 1 engine discharge: 0.8MPa, suction: 0.4MPa | |
| HFO pre-boost press | nominal | 0.4MPa |
| HFO press. before engine | nominal | 0.7MPa |
| HFO press. before fuel inj. pump | operation | 0.5–1.0MPa |
| | Low alarm | 0.35MPa |
| HFO press. after engine | nominal | 0.4MPa |
| MDO press. before feed pump | nominal | static |
| MDO press. after pump relief valve | nominal | static |
| MDO press. before engine | nominal | 0.8MPa |
| MDO press. after engine | nominal | 0.4MPa |
| HFO viscosity before fuel inj. pump | operation | 10-14mm²/s (cSt) |
| MDO viscosity before fuel inj. pump | operation | ≥2.0mm²/s (cSt) |
| HFO filter before engines | auto-backwash type | |
| (To be installed on the circulation line) | nominal | 10micron |
| | Recommend to install it on the circulation line. in case it is on the supply line, 350 mesh strainer should be installed on the circulation line. | |
| Bypass strainer | mesh size | 350 mesh |
| (For HFO auto-backwash filter maintenance) | | |
| FO pump suction strainer | mesh size | 60-100 mesh |
| (For each pump installed into hull side system) | | |

## 6. COMPRESSED AIR SYSTEM

6.1 Engine fitting (each 1 set/engine)

| | |
|---|---|
| Starting operation button | for manual start at engine-side<br>Electrical push button type |
| Main starting air valve | starting air control to engine with magnetic valve(88V) |
| Starting air rotary valve | for the pilot air distribution to starting valve |

6.2 Separate accessories (installed on hull part by shipyard):None

6.3 Reference data for External system

| | | |
|---|---|---|
| Starting air source | nominal | 3.0MPa |
| | operation | 1.5-3.0MPa |
| | low alarm | 1.5MPa |
| Starting air consumption (for manual start) | 0.3 $m^3$ normal/one (1) time (At engine side) | |
| Control air pressure | operation | 0.7-0.9MPa |
| Control air consumption | negligible small | |
| Auxiliary air reservoir capacity | 100L × 3.0MPa(1 Bottle/ship) | |

Remarks: It is necessary to be supplied control air for dead ship start.
Control air for dead ship start is to be arranged by Shipyard.

## 7. LUBRICATING SYSTEM

7.1 Engine fitting (each 1 set/engine)

| | |
|---|---|
| L.O suction strainer(for pump inlet) | punching board type of 3mm holes built in LO tank |
| L.O. pump | engine driven gear type<br>19.0$m^3$/h x 0.6MPa |
| L.O. bypass filter | centrifugal glacier type<br>Model: FM200 |
| L.O. cooler | plate type<br>fresh water cooled |
| L.O. thermostatic valve | wax type with hand operating bypass device |
| L.O relief valve | bearing line setting pressure 0.4-0.5 MPa |
| L.O. filter | continuous auto-backwash type<br>absolute 30μm<br>with differential pressure gauge and switch |
| L.O. tank | incorporated in common bed with overflow port<br>capacity abt. 1,100lit |
| L.O. priming pump | motor driven gear rotary type<br>pump 2.5$m^3$/h × 0.2MPa<br>motor 0.94kW AC 440V 60Hz 3-phase<br>TEFC(IP44) class F insulation |

7.2 Separate accessories (installed on hull part by shipyard): None

7.3 Reference data for External system

| | | |
|---|---|---|
| Bearing L.O. pressure | operation | 0.4-0.5MPa |
| | low alarm | 0.35MPa |
| | safety stop | 0.30MPa |
| Turbocharger L.O. pressure | operation | 0.06-0.15MPa |
| L.O. temperature of engine inlet | operation | 50-60°C |
| | High alarm | 65°C |
| L.O. centrifugal separator flow (Overflow tank continuous purification system) | min. flow 880lit/h for 3 engines | |
| Treatment temp. for separator | 90-95°C | |
| Overflow tank | Capacity ≥1,000L per one (1) ship (shipyard supply) | |

## 8. COOLING WATER SYSTEM

8.1 Engine fitting (each 1 set/engine)

| | |
|---|---|
| Cylinder CFW pump | engine driven centrifugal type<br>30m³ /h×18mTH |
| Cooler CFW pump | engine driven centrifugal type<br>30m³ /h×18mTH |
| Engine warming port | 5K-25A |
| Air vent outlet for cylinder CFW | 5K-15A with valve |
| Jacket CFW temp.control valve | wax type with hand operating bypass device<br>Engine outlet control: 75-85°C |

8.2 Separate accessories (installed on hull part by shipyard): Supply by Yard

| | |
|---|---|
| LT-FW Cooler | plate type with temperature control valve |
| Engine pre-heating unit | electric type FW heater with control panel (IP44)<br>heater capacity 20kW, FW pump 6.0m³/h×10m,<br>power source AC440V, 60Hz, 3phase<br>(1unit/ship) |

8.3 Reference data for External system

| | | |
|---|---|---|
| FW expansion tank height | tank bottom higher 3–10m than crankshaft center | |
| FW expansion tank net capacity | min.240lit per 1 ship | |
| Cooler CFW temp. of engine inlet | 38°C | |
| Cooler CFW pressure | operation | max 0.40 MPa |
| Engine coolers flow resistance | approx.. 8mAq | |
| Cylinder CFW pressure of cyl. inlet | operation | 0.15-0.4MPa |
| | low alarm | 0.15MPa |
| Cylinder CFW flow resistance | approx. 6mAq | |
| Cylinder CFW temp. of engine outlet | high alarm | 90°C |
| | safety stop | 95°C |

## 9. SAFETY DEVICE

9.1 Engine fitting (each 1 set/engine)

| | |
|---|---|
| Indicator valve | equipped on each cylinder head<br>with Max. pressure gauge connection |
| Crankcase relief valve | equipped properly on the side-cover |

## 10. SUNDRY

10.1 Engine fitting (each 1 set/engine)

| | |
|---|---|
| Step | on FO injection pump side |
| Flywheel turning device | hand operating bar type |
| Flywheel cover | steel plate fabrication |
| Exhaust manifold cover | steel plate fabrication with inside lagging |
| Common bed | welded steel plate fabrication |
| Jack bolts | screwed into the common bed |

10.2 Separate accessories (installed on hull part by shipyard): None

**11. ENGINE GAUGE BOARD** Engine fitting consists of following indicators

Engine tachometer ..... direct measuring mechanical type
Pressure gauge..... direct measuring bourdon tube type glycerin-filled type
each one(1) of; Charge air
Fuel oil
LO
T/C LO
Boost air
Cylinder CFW
Cooler CFW
Starting air

**12. LOCAL THERMOMETER** Engine fitting of direct measuring liquid filled glass stem type

Charge air.....intake manifold
Exhaust gas .....each cylinder outlet
turbo-charger in/outlet
Fuel oil.....engine inlet
Lube oil.....engine inlet
LO cooler inlet
Cylinder CFW engine inlet
engine common outlet
Cooler CFW .....air cooler outlet
LO cooler inlet

## 13. CONTROL SYSTEM

13.1 Engine fitting (each 1 set/engine)

| | |
|---|---|
| Starting magnetic valve | DC24V 16W |
| Fuel control magnetic valve | DC24V 16W<br>for fuel control in process of starting |
| Remote stop magnetic valve | DC24V 16W<br>for the fuel shutdown device of control system and safety system |
| F.O. injection pump fuel cut magnetic valve | DC24V 16W<br>for each cylinder fuel cut devices of safety system |
| Handle switch | for remote start stand-by(HS) |
| Turning safety switch | for start interlock(TS) |
| Speed sensor | pulse sensing type for engine speed |
| Speed switch unit | signal output for Low-speed and Over-speed installed into DECSS |
| LO press. switch (automatic stop) | 0.3MPa(63Q2) |
| Over-speed switch (automatic stop) | 112–115% by Speed switch unit(12) |
| Cylinder CFW temp. switch (automatic stop) | 95°C(26WH2) |
| LO sump tank level switch (L. alarm) | too low(33QL) |
| LO auto-backwash filter differential press. switch (H. alarm) | preset 0.09MPa(63QDM) |
| LO T/C inlet press. switch(L alarm) | 0.06MPa below(63QT) |
| FO inj. pipe leakage level switch (H. alarm) | high level alarm in the leakage box(33F) |
| Engine condition display | engine fitted type<br>Power source DC24V<br>(supply engine control unit) |
| Engine control unit (IP44) | engine fitted type<br>Power source : main supply DC-24V<br>Back up supply DC24V<br>(battery or UPS)<br>Power consumption: Rated 1.8A / DC24V<br>Max. 5.0A / DC24V<br>Connection link: RS422 (Modbus-RTU) |
| Starter panel for LO priming pump | engine fitted type<br>AC440V 3-phase 60Hz |

13.2 Separate accessories (installed on hull part by shipyard):None

## 14. MONITORING SYSTEM

14.1 Engine fitting

14.1.1 Pressure transmitter DC 24V output 4 - 20mA in proportion to the scale range

| measuring point | Scale range | Normal operation | alarm setting | q'ty/eng |
|---|---|---|---|---|
| L.O. engine inlet(PTQ) | 0–1.0MPa | 0.4–0.5MPa | 0.35MPa below | 1 |
| Cyl.CFW engine inlet(PTWH) | 0–0.6MPa | 0.15–0.40MPa | 0.15MPa below | 1 |
| Cyl.Cooler CFW engine inlet(PTWL) | 0–0.6MPa | 0.15–0.40MPa | 0.15MPa below | 1 |
| F.O. engine inlet(PTF) | 0-1.6MPa | 0.5-1.0MPa | 0.35MPa below | 1 |
| Starting air inlet(PTAS) | 0-4.0MPa | 1.5-3.0MPa | 1.5MPa below | 1 |

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14.1.2 Thermo-resistance bulb Pt100 R100/R0=1.3850
Indication value of temp.sensor might be different from local thermometer due to different fitting location.

| measuring point | Scale range | Normal operation | alarm setting | q'ty |
|---|---|---|---|---|
| LO engine inlet (TRQ) | max. 200°C | 50 – 60°C | 65°C over | 1 |
| Cyl.CFW engine outlet(TRWH) | max. 200°C | 75-85°C | 90°C over | 1 |
| Exh.gas T/C inlet(TRTI) | max. 700°C | ＜580°C | 580°C over | 2 |
| Exh.Gas each cylinder outlet(TRC) | max. 700°C | ＜480°C | 480°C over | 6 |
| Exh T/C outlet(TRTO) | max. 700°C | ＜480°C | 480°C over | 1 |

14.2 Separate accessories (installed on hull part by shipyard): None

## 15. SPARE PARTS & TOOLS

Spare parts and tools will be supplied in accordance with Daihatsu standard practice which is satisfied the requirement of classification society.
Details will be described in "SPARE PARTS LIST" and "TOOLS LIST" for approval plan.

## 16. SHOP TRIAL

The shop trial should be performed in accordance with requirements of classification society. The diesel generator engine should be run to couple up a generator unless particular reasons, and to use Marine Diesel Oil for fuel. The load facility is of water resistance.

16.1 Test & inspection items

16.1.01 Measurement of Crank deflection
Crank deflection should be measured in cold condition before running, and in hot condition after running also measured(Only No.1 Cylinder)

16.1.02 Starting test
The number of starting times and minimum starting air pressure will be recorded by using of an accessory air reservoir or shop's one, testing from cold condition.

16.1.03 Load test
Under each load rate described afterward, engine data such as fuel consumption, pressure, temperature should be measured and recorded.

16.1.04 Measurement of Bearing temperature
After running, Main bearing and Crankpin bearing should be taken temperature.

16.1.05 Working test for protecting device
In case that the remote control panel is in our scope, its working should also be demonstrated.

16.1.06 Generating characteristics test
Voltage and frequency should be measured and recorded when generator output (kW) is changed in order of 100%, 75%, 50%, 25%, 50%, 75%,100%,110% and 100%.

16.1.07 Governor test
Momentary and permanent speed variations and the speed recovery time to steady speed should be confirmed to be controlled within requirements of classification and recorded, when full load is suddenly taken off or, when load is suddenly applied with loading steps approved by classification society. Speed variations are measured by the frequency meter.

16.1.08 Parallel running test
After setting the load(kW) to each generator to 75%, each generator output (kW) will be measured and recorded when total combine load is changed in order of 75%,100%,80%,60%,40% 20%, 40%, 60%,80%100%75%.

16.1.09 Manual voltage change test
Using the voltage control rheostat which is supply by generator maker, the range of voltage control will be recorded.

16.1.10 Frequency change with governor motor
Frequency change speed will be confirmed to be 7-10 s/Hz and recorded.

16.1.11 Remote starting test
In case that the remote control panel is in our scope of supply, remote starting test will be demostarated.

16.1.12 Measurement of torsional vibration

The previous measured data of the same diesel generator installation will be submitted. If it were not for such data, the torsional vibration should be measured only one(1) engine of the first ship.

16.2 Open up inspection

One(1) cylinder per ship will be opened up and items of its cylinder should be inspected as below;

-Cylinder cover
-Intake & Exhaust valve
-Piston(Piston ring will be fitted as it standard)
-Connection rod
-Crankpin bearing
-Crankpin bolt

One(1) main bearing per ship will be opened up and inspected.

16.3 Attendant schedule

X: attend

| Test & Inspection items | | Customer witness | Manufacturer |
|---|---|---|---|
| 1)Measure of crank deflection | | | X |
| 2)Starting test | 25% x 20 min | | X |
| 3)Generator load(kW) test Including the fuel consumption Measurement at each load | 50% x 20 min | | X |
| | 75% x 20 min | | X |
| | 100% x 60 min | X | X |
| | 110% x 30 min | | X |
| 4)Measure of bearing temperature | | | X |
| 5)Working test of protective device | Emergency trip | X | X |
| | Pre-alarm | | X |
| 6)Generator characteristic test | | | X |
| 7)Governor test | | | X |
| 8)Parallel running test | | | X |
| 9)Manual voltage change test | | | X |
| 10)Frequency change test with governor motor | | | X |
| 11)Remote starting test with remote control panel | | X | X |
| 12)Measure of torsional vibration | | | X |
| All of above data should be submitted to customer. | | | |

16.4 Witness saving

If customer accept it, the manufacturer will not request the customer's witness shop trial. In that case, the manufacturer should take responsibility of doing the shop trial and the presentation of trial record as a matter of course.